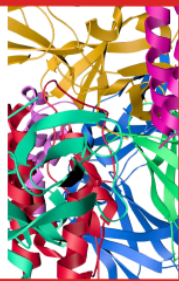


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Reconstitution of Circulating Mucosal-Associated Invariant T Cells after Allogeneic Hematopoietic Cell Transplantation: Its Association with the Riboflavin Synthetic Pathway of Gut Microbiota in Cord Blood Transplant Recipients

Takaaki Konuma,* Chisato Kohara,* Eri Watanabe,[†] Shunsuke Takahashi,[‡] Genki Ozawa,[‡] Kei Suzuki,[‡] Motoko Mizukami,[§] Etsuko Nagai,[§] Koji Jimbo,* Yuta Kaito,* Masamichi Isobe,* Seiko Kato,* Satoshi Takahashi,* Asako Chiba,[¶] Sachiko Miyake,[¶] and Arinobu Tojo*

Mucosal-associated invariant T (MAIT) cells are a type of innate lymphocyte and recognize riboflavin (vitamin B2) synthesis products presented by MHC-related protein 1. We investigated long-term reconstitution of MAIT cells and its association with chronic graft-versus-host disease (cGVHD) in a cross-sectional cohort of 173 adult patients after allogeneic hematopoietic cell transplantation (CBT) but not with time after bone marrow transplantation or peripheral blood stem cell transplantation. The number of MAIT cells was significantly lower in patients with cGVHD compared with patients without cGVHD. We also examined the association between MAIT cell reconstitution and gut microbiota as evaluated by 16S ribosomal sequencing of stool samples 1 mo post-CBT in 27 adult patients undergoing CBT. The diversity of gut microbiota was positively correlated with better MAIT cell reconstitution after CBT. Phylogenetic Investigation of Communities by Reconstruction of Unobserved States analysis indicated that amounts of *ribB* and *ribA* genes were significantly higher in the microbiomes of patients with subsequent MAIT cell reconstitution after CBT. In conclusion, long-term MAIT cell reconstitution is dependent on the type of donor source. Our data also unveiled an important role for the interaction of circulating MAIT cells with gut microbiota in humans. *The Journal of Immunology*, 2020, 204: 1462–1473.

Allogeneic hematopoietic cell transplantation (HCT) is an established curative therapy for patients with hematological disorders but causes unique morbidities, such as graft rejection (host-versus-graft reaction), graft-versus-host disease (GVHD), and opportunistic infections associated with interactions between donor and recipient immune cells and administration of immunosuppressive agents (1, 2). However, the mechanisms of these immunological pathologies have not been established, although alloreactive conventional T cells, B cells, and other immune effector cells may be involved in development of GVHD.

Mucosal-associated invariant T (MAIT) cells are innate lymphocytes comprising 1–10% of CD3⁺ cells in humans and

are abundant not only in mucosal tissues but also in peripheral blood (PB) (3, 4). MAIT cells express semi-invariant TCRs Vα7.2-Jα33 (TRAV1-2-TRAVJ33) and the C-type lectin-like receptor CD161 in humans. MAIT cells recognize riboflavin (vitamin B2) synthesis products presented by the nonpolymorphic MHC class I-like molecule MHC-related protein 1 (MR1) (5, 6).

MAIT cell expansion requires commensal microbiota and circulating B cells in mouse models (3, 7, 8) and in humans (7, 9). Further, Bhattacharyya et al. (9) also demonstrated that increased gut abundance of *Blautia* species and *Bifidobacterium longum* was correlated with better circulating MAIT cell reconstitution after allogeneic PB stem cell transplantation (PBSCT). However, because

*Department of Hematology/Oncology, The Institute of Medical Science, The University of Tokyo, Tokyo 108-8639, Japan; [†]Clinical Flow Cytometry Laboratory, The Institute of Medical Science, The University of Tokyo, Tokyo 108-8639, Japan; [‡]TechnoSuruga Laboratory Co., Ltd., Shizuoka 424-0065, Japan; [§]Department of Laboratory Medicine, The Institute of Medical Science, The University of Tokyo, Tokyo 108-8639, Japan; and [¶]Department of Immunology, Juntendo University School of Medicine, Tokyo 113-8421, Japan

ORCID: 0000-0002-1953-8463 (T.K.); 0000-0002-1076-2338 (Satoshi Takahashi); 0000-0001-8861-0644 (A.C.); 0000-0001-9016-1948 (A.T.).

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T.K. conceived the project, designed the research, performed experiments of MAIT cells, analyzed data, and wrote the paper. C.K. and E.W. also performed experiments of MAIT cells. Shunsuke Takahashi, G.O., and K.S. performed experiments and data analysis of microbiome. A.C. and S.M. supported experiments of MR1 tetramer assay. All the other authors participated in the treatment of the

patients, acquired the clinical data, and contributed to writing the paper. All authors approved the final version.

Address correspondence and reprint requests to Dr. Takaaki Konuma, Department of Hematology/Oncology, The Institute of Medical Science, The University of Tokyo, 4-6-1, Shirokanedai, Minato-ku, Tokyo 108-8639, Japan. E-mail address: tkonuma@ims.u-tokyo.ac.jp

The online version of this article contains supplemental material.

Abbreviations used in this article: aGVHD, acute GVHD; BMT, bone marrow transplantation; CBT, cord blood transplantation; cGVHD, chronic GVHD; 6-FP, 6-formylpterin; GVHD, graft-versus-host disease; HCT, hematopoietic cell transplantation; IM, ionomycin; KEGG, Kyoto Encyclopedia of Genes and Genomics; MAIT, mucosal-associated invariant T; MR1, MHC-related protein 1; MSD, matched sibling donor; NIH, National Institutes of Health; 5-OP-RU, 5-(2-oxopropylideneamino)-6-D-riboflavinouracil; PB, peripheral blood; PBSCT, PB stem cell transplantation; PD-1, programmed cell death-1; PICRUSt, Phylogenetic Investigation of Communities by Reconstruction of Unobserved States; SEA, staphylococcal enterotoxin A; UCB, unrelated cord blood.

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cord blood grafts contain lower numbers of MAIT cells than adult donor sources, the effects of gut microbiota and circulating B cells on MAIT reconstitution after cord blood transplantation (CBT) in clinical settings remain unknown.

MAIT cells migrate to inflamed tissues and have the capacity to produce several cytokines, including IFN- γ and IL-17, and release cytotoxic granules, such as granzyme and perforin, in MR1-dependent and -independent manners (10, 11). Several studies have reported a role for MAIT cells in various autoimmune and inflammatory diseases (12, 13), including acute GVHD (aGVHD) in mice (14) and humans (9, 15, 16). However, the potential dysregulation of MAIT cells in patients with chronic GVHD (cGVHD) is incompletely understood (17). Therefore, we examined long-term reconstitution of MAIT cells after allogeneic HCT and the relationship between MAIT cells and cGVHD. We used a cross-sectional approach because of the difficulty of performing a longitudinal study in long-term follow-up after allogeneic HCT. We also examined the association between short-term MAIT cell reconstitution, gut microbiota, circulating B cells, and the riboflavin synthesis pathway of gut microbiota in adult patients undergoing CBT.

Materials and Methods

Patient population and sample collection

The present study was conducted at the Institute of Medical Science of the University of Tokyo. In the cross-sectional cohort, 173 patients at our institute who had received outpatient treatment between February 2015 and February 2018 and had not experienced disease recurrence for at least 12 mo after undergoing allogeneic HCT were included (Table I). In contrast, to clarify the short-term reconstitution of MAIT cells after unrelated CBT, we retrospectively analyzed reconstitution of MAIT cells in 27 patients that received CBT at our institute between September 2014 and January 2017 (Table II), after they were prospectively enrolled in PB and fecal collection studies approved by the Institutional Review Board (27-19-0713, 29-90-B0328). Patients were included if their PB samples were evaluated for 12 mo after CBT without disease recurrence. Samples from 15 healthy subjects and patients without hematological disease were used as control samples. The median age of the control groups was 54 y (range, 35–83 y). Six of the control groups were male, and nine were female. cGVHD was assessed and graded at the time of sample collection according to the 2014 National Institutes of Health (NIH) consensus criteria (18). Written informed consent was acquired from all patients and healthy subjects.

Flow cytometric analysis

Complete blood counts were conducted using an automated hematology analyzer XE-2100 (Sysmex, Kobe, Japan). The proportion of lymphocytes was quantified using manual differential leukocyte counts. PBMCs were separated from PB samples by density gradient medium centrifugation using a Lymphoprep kit (Axix-Shield PoC, Oslo, Norway) according to the manufacturer's instructions and stored in cell freezing medium (Bambanker; Nippon Genetics, Tokyo, Japan) until analysis.

To examine the proportion of MAIT cells, PBMCs were stained with allophycocyanin-Cy7-conjugated anti-human CD3 (clone SK7; BioLegend, San Diego, CA), FITC-conjugated anti-human TCR γ/δ (clone B1; BioLegend), PE-conjugated anti-human CD161 (clone HP-3G10; Tonbo Biosciences, San Diego, CA), BV421-conjugated anti-human TCR V α 7.2 (clone 3C10; BioLegend), BV510-conjugated anti-human CD4 (clone RPA-T4; BioLegend), PE-Cy7-conjugated anti-human CD8 α (clone RPA-T8; Tonbo Biosciences), allophycocyanin-conjugated anti-human programmed cell death-1 (PD-1) (clone EH12.2H7; BioLegend), and PerCP/Cy5.5-conjugated anti-human TCR V α 24-J α 18 (clone 6B11; BioLegend) Abs. The gating strategy for MAIT cells is described in a previous report (3) (Fig. 1A). MAIT cells were defined as the expression of CD161 and TCR V α 7.2 in CD3⁺TCR γ/δ ⁺ cell populations. Around 10,000–50,000 events of CD3⁺TCR γ/δ ⁺ cells were collected and analyzed to determine the population of CD161⁺TCR V α 7.2⁺MAIT cells. Flow cytometry was performed using a FACS Aria II (BD Biosciences, San Jose, CA), and data were analyzed using FlowJo 10.3 software (FlowJo LLC, Ashland, OR).

MR1 tetramer assay

PBMCs were also stained with BV421-conjugated human MR1:5-(2-oxopropylideneamino)-6-D-ribitylamino-uracil (5-OP-RU) tetramer or human MR1:6-formylpterin (6-FP) tetramer, which was obtained from the NIH Tetramer Core Facility (6). For tetramer staining, cells in suspension were incubated with MR1: 5-OP-RU tetramer for 40 min at room temperature followed by other surface stains for 20 min, according to facility's instructions.

In vitro functional analysis

CD161 expression in T cells is a marker of type-17 differentiation program with specific tissue homing properties and mainly includes IL-17-producing CD4⁺Th17 cells and IL-17-producing CD8⁺Tc17 cells (19). The latter is known to consist mainly of MAIT cells. Because we did not have a sufficient number of cells to perform a MAIT cell functional assay in patients after allogeneic HCT, we used CD161⁺T cells and conventional CD4⁺ and CD8⁺T cells for in vitro functional analysis. To examine cytokine production in conventional CD4⁺ and CD8⁺T cells and CD4⁺CD161⁺ and CD8⁺CD161⁺T cells, including the MAIT cell subset, PBMCs were incubated in RPMI 1640 medium with 10% FBS, 100 IU/ml penicillin, and 100 μ g/ml streptomycin for 6 h in the presence or absence of PMA, ionomycin (IM), brefeldin A, and monensin using the Cell Stimulation Cocktail (Tonbo Biosciences); 10 ng/ml IL-12 p70 (Tonbo Biosciences), 10 ng/ml IL-15 (PeproTech), and/or 10 ng/ml IL-18 (R&D systems, Minneapolis, MN); or 20 ng/ml staphylococcal enterotoxin A (SEA). PBMCs were then fixed and permeabilized using a BD Cytotfix/Cytoperm Fixation/Permeabilization Kit (BD Biosciences) and stained with PerCP-Cy5.5-conjugated anti-human CD3 (clone UCHT-1; BioLegend), BV510-conjugated anti-human CD4, PE-Cy7-conjugated anti-human CD8 α , FITC-conjugated anti-human CD161, Alexa Fluor 647-conjugated anti-human granzyme B (clone GB11; BioLegend), PE-conjugated IFN- γ (clone 4S.B3; BioLegend), BV421-conjugated anti-human IL-17A (clone BL168; BioLegend), and allophycocyanin/Fire 750-conjugated anti-human CD69 (clone FN50; BioLegend) Abs. Dead cells were excluded using Ghost Dye Red 710 (Tonbo Biosciences), according to manufacturer's instructions.

To examine altered expression of master transcription factors in conventional CD4⁺ and CD8⁺T cells and CD4⁺CD161⁺ and CD8⁺CD161⁺T cells, including the MAIT cell subset, PBMCs were incubated for 6 h in the presence or absence of PMA, IM, brefeldin A, and monensin using the Cell Stimulation Cocktail; or 10 ng/ml IL-12 p70, 10 ng/ml IL-15, and 10 ng/ml IL-18; or 20 ng/ml SEA. Cells were then fixed and permeabilized and stained with allophycocyanin-Cy7-conjugated anti-human CD3 (clone UCHT-1; BioLegend), BV510-conjugated anti-human CD4, PE-Cy7-conjugated anti-human CD8 α , FITC-conjugated anti-human CD161, BV421-conjugated anti-human ROR γ t (clone Q21-559; BD Bioscience), Alexa Fluor 647-conjugated PLZF (clone R17-809; BD Bioscience), PE-conjugated anti-human T-bet (clone 4B10; BioLegend), and PerCP/Cy5.5-conjugated anti-human GATA3 (clone 16E10A23; BioLegend) Abs. Dead cells were excluded using Ghost Dye Red 710 (Tonbo Biosciences), according to manufacturer's instructions.

Gut microbiota sequencing

To examine the association between gut microbiota and MAIT cell reconstitution after CBT, fecal samples were collected from 27 patients undergoing CBT in our institute at ~1 mo after CBT. Bacterial genomic DNA in fecal specimens was extracted from bead-treated suspensions using Magstrat System magLEAD 12gC and magLEAD series MagDEA DNA 200 (Precision System Science, Chiba, Japan). The V3-V4 region of 16S rRNA-encoding DNA was amplified using Pro341F and Pro805R primer sets with sample-specific 8-bp barcode sequences (20, 21). Next-generation sequencing analysis was conducted using a paired end and modified to a 2 $\times$ 281-bp cycle run on an Illumina MiSeq sequencing system and MiSeq Reagent Kit version 3 (600 cycle). After sequencing was complete, 281-bp dual-indexed paired-end reads were first joined by aligning overlapping sequences with fastq-join quality, and quality filtering was performed as previously described (20, 21). Only reads that had quality value scores of >20 for >99% of the sequence were extracted for further analysis (20). Chimeric sequences were removed using USEARCH 6.1 (22). Operational taxonomy units were selected using an open-reference approach with the GreenGenes 13.8 database (23). The resulting sequence data were processed with QIIME version 1.8.0 (24).

Metagenomic prediction

To evaluate the effect of putative metabolic function of gut microbiota on MAIT cell reconstitution after CBT, Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) version 1.1.1

analysis (25) was used to infer metagenomes based on 16S marker data and predicted Kyoto Encyclopedia of Genes and Genomics (KEGG) pathway abundances (26). Statistical analysis and visualization of KEGG pathway data were performed with Statistical Analysis of Metagenomic Profiles version 1.1.3 software using a Welsh *t* test (27).

Statistical analysis

All statistical analyses were performed using GraphPad Prism 6 for Mac OS X (GraphPad Software, San Diego, CA), R 3.1.0 software (R Foundation for Statistical Computing, Vienna, Austria), and QIIME 1.8.0 (24). The bar indicated mean values. Data were expressed as the mean $\pm$ SD. Statistical significance was calculated by using the unpaired Student *t* tests among two groups. Statistical significance was calculated by using the one-way ANOVA followed by Bonferroni post hoc comparisons among three or more groups. Correlations between continuous variables were examined using linear regression analysis. All *p* values were two sided, and *p* values <0.05 were considered statistically significant.

Results

Characteristics of patients and transplantation

The characteristics of patients and transplantations in the cross-sectional cohort are shown in Table I. The median age of the patients was 49 y (range, 18–68 y). The most common disease type was acute myelogenous leukemia (34%), and the most common graft source was unrelated cord blood (UCB) (70%). Myeloablative conditioning regimens (93%) and cyclosporine and methotrexate for GVHD prophylaxis (79%) were commonly performed. At the time of PB sample collection, 98 patients

(57%) had cGVHD, and 29 patients (17%) had received immunosuppressive treatment. The median period after HCT was 66 mo (range, 12–376 mo). There were no significant differences in baseline characteristics among donor sources, except for disease type ($p = 0.014$), conditioning regimen ($p < 0.001$), GVHD prophylaxis ($p < 0.001$), and median time after HCT ($p < 0.001$) (Table I).

MAIT cell reconstitution in long-term survivors after allogeneic HCT

Among patients who had undergone allogeneic HCT >1 y ago, we evaluated long-term reconstitution of MAIT cells in cross-sectional cohort of 173 patients after allogeneic HCT (Table I). The proportions of MAIT cells among CD3⁺T cells and numbers of MAIT cells were significantly lower in patients that received HCT from a matched sibling donor (MSD) and UCB compared with the control group (Fig. 1B). However, there were no significant differences in the proportion of NKT cells and γ/δ T cells among CD3⁺T cells between the control group and patients that received HCT from MSD, matched unrelated donor, or UCB (Fig. 1C). The proportion of MAIT cells among CD3⁺T cells was significantly associated with the proportion of CD8⁺CD161⁺T cells (Fig. 1D). The number of MAIT cells was significantly associated with the number of CD4⁺CD161⁺T cells and CD8⁺CD161⁺T cells (Fig. 1E). In addition to the proportion of MAIT cells, the proportion of CD8⁺CD161⁺T, but not CD4⁺CD161⁺T, cells was significantly lower in patients that received HCT from MSD and UCB compared

Table I. Characteristics of patients and transplantations

Characteristic	Total	Related BMT/PBSCT	Unrelated BMT	Unrelated CBT	<i>p</i> Value
Number of patients	173	38	14	121	
Median age at study in y (range)	49 (18–68)	51 (31–68)	50.5 (26–68)	49 (18–68)	0.616
Sex, <i>n</i> (%)					
Male	89 (51.4)	15 (39.5)	5 (35.7)	69 (57.0)	0.079
Female	84 (48.6)	23 (60.5)	9 (64.3)	52 (43.0)	
Disease type, <i>n</i> (%)					
AML	59 (34.1)	11 (28.9)	1 (7.1)	47 (38.8)	0.014
ALL	43 (24.9)	6 (15.8)	5 (35.7)	32 (26.4)	
MDS	43 (24.9)	8 (21.1)	5 (35.7)	30 (24.8)	
CML/MPN/CMML	16 (9.2)	6 (15.8)	2 (14.3)	8 (6.6)	
NHL	4 (2.3)	1 (2.6)	0	3 (2.5)	
MM	1 (0.6)	1 (2.6)	0	0	
SAA	7 (4.0)	5 (13.2)	1 (7.1)	1 (0.8)	
Conditioning regimen, <i>n</i> (%)					<0.001
MAC	161 (93.1)	29 (76.3)	13 (92.9)	119 (98.3)	
RIC	10 (5.8)	7 (18.4)	1 (7.1)	2 (1.7)	
Unknown	2 (1.2)	2 (5.3)	0	0	
GVHD prophylaxis					<0.001
CSP + MTX	136 (78.6)	30 (78.9)	4 (28.6)	102 (84.3)	
CSP + MMF	14 (8.1)	1 (2.6)	0	13 (10.7)	
CSP	10 (5.8)	4 (10.5)	0	6 (5.0)	
TAC + MTX	9 (5.2)	0	9 (64.3)	0	
TAC + MMF	1 (0.6)	0	1 (7.1)	0	
Unknown	3 (1.7)	3 (7.9)	0	0	
Median time after HCT in mo (range)	66 (12–376)	211.5 (12–376)	96.5 (13–251)	46 (12–187)	<0.001
cGVHD, <i>n</i> (%)					0.155
None	75 (43.4)	20 (52.6)	8 (57.1)	47 (38.8)	
Mild	54 (31.2)	8 (21.1)	3 (21.4)	43 (35.5)	
Moderate	33 (19.1)	6 (15.8)	1 (7.1)	26 (21.5)	
Severe	11 (6.4)	4 (10.5)	2 (14.3)	5 (4.1)	
Systemic immunosuppressive treatments, <i>n</i> (%)					0.119
None	144 (83.2)	34 (89.5)	10 (71.4)	100 (82.6)	
Calcineurin inhibitors	11 (6.4)	1 (2.6)	1 (7.1)	9 (7.4)	
Glucocorticoids	9 (5.2)	2 (5.3)	0	7 (5.8)	
Calcineurin inhibitors + glucocorticoids	9 (5.2)	1 (2.6)	3 (21.4)	5 (4.1)	

ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CML, chronic myelogenous leukemia; CMML, chronic myelomonocytic leukemia; CSP, cyclosporine; MAC, myeloablative conditioning; MDS, myelodysplastic syndrome; MM, multiple myeloma; MMF, mycophenolate mofetil; MPN, myeloproliferative neoplasm; MTX, methotrexate; NHL, non-Hodgkin lymphoma; RIC, reduced-intensity conditioning; SAA, severe aplastic anemia; TAC, tacrolimus.

The *p* values in bold are statistically significant (< 0.05).

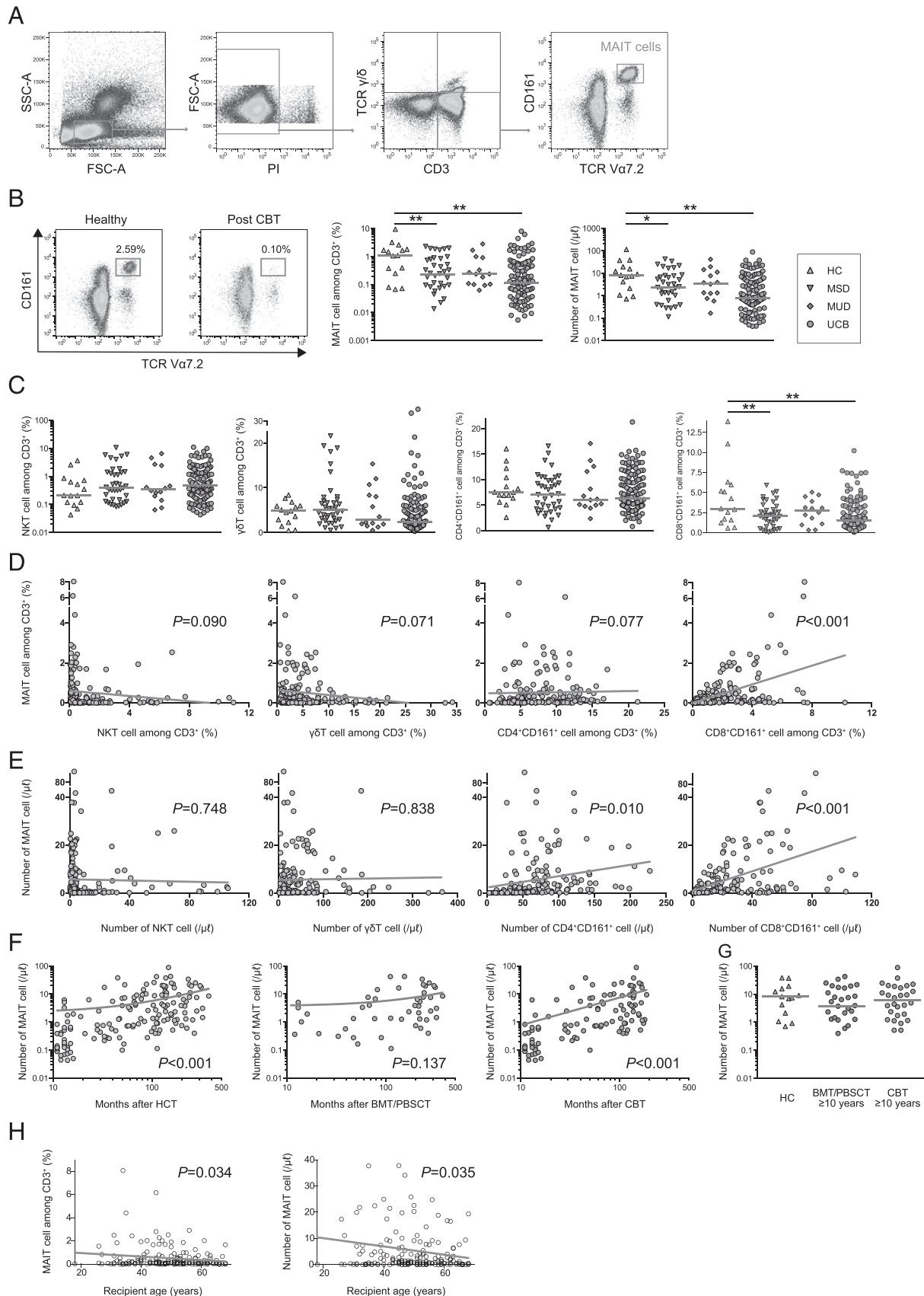


FIGURE 1. Reconstitution of circulating MAIT cells in long-term survivors after allogeneic HCT ($n = 173$). **(A)** Gating strategies for MAIT cells, **(B)** proportions, and numbers of MAIT cells after allogeneic HCT and age-matched healthy controls (HC). **(C)** Proportions of NKT cells, $\gamma\delta$ T cells, CD4⁺CD161⁺T cells, and CD8⁺CD161⁺T cells among CD3⁺T cells according to donor source. **(D)** Correlation of proportion of MAIT cells and NKT cells, $\gamma\delta$ T cells, CD4⁺CD161⁺T cells, and CD8⁺CD161⁺T cells among CD3⁺T cells. **(E)** Correlation of number of MAIT cells and NKT cells, $\gamma\delta$ T cells, CD4⁺CD161⁺T cells, and CD8⁺CD161⁺T cells. **(F)** Correlation between number of MAIT cells and months after allogeneic BMT/PBSCT, and CBT. **(G)** Number of MAIT cells over 10 y after BMT/PBSCT and CBT and age-matched HC. **(H)** Correlation (Figure legend continues)

with the control group (Fig. 1C). The number of MAIT cells significantly increased with months after HCT (Fig. 1F). According to donor source, the number of MAIT cells was significantly correlated with months after CBT but not months after bone marrow transplantation (BMT)/PBSCT (Fig. 1F). There were no significant differences in the number of MAIT cells between the control group and patients beyond 10 y after BMT/PBSCT and CBT (Fig. 1G). Moreover, the proportion and number of MAIT cells were significantly correlated with recipient age (Fig. 1H).

MAIT proportions in relation to cGVHD

Next, we evaluated the correlation between MAIT cells and cGVHD. The number of MAIT cells was significantly higher in patients with mild cGVHD compared with patients without cGVHD (Fig. 2A). Contrary to previous findings that MAIT cells were predominantly CD8⁺ in patients without cGVHD and in healthy controls (3, 4), the proportion of CD4⁺MAIT cells was significantly higher and the proportion of CD8⁺MAIT cells was significantly lower in patients with mild and moderate/severe cGVHD compared with patients without cGVHD (Fig. 2C), which indicated that MAIT cell CD4/CD8 ratios were slightly higher in patients with moderate/severe cGVHD compared with patients without cGVHD (Fig. 2B), but this was NS.

PD-1 is an inhibitory receptor expressed on activated and exhausted T cells (28) and is associated with the development of cGVHD (29). Therefore, we evaluated expression levels of PD-1 in MAIT cells. There were no significant differences in MAIT cell PD-1 expression levels and proportions of PD-1⁺ MAIT cells between patients with and without cGVHD (Fig. 2D). In addition, no significant differences were observed in PD1 expression levels in CD4⁺CD161⁺T cells and CD8⁺CD161⁺T cells between patients with and without cGVHD (Fig. 2D).

Immunosuppressive treatments may have affected the proportions and numbers of MAIT cells in patients who developed or were at risk for cGVHD. Therefore, we further investigated the association between MAIT cells and administration of calcineurin inhibitors or glucocorticoids. Administration of calcineurin inhibitors and glucocorticoids was not significantly associated with the proportions and numbers of MAIT cells (Fig. 2E, 2F), indicating that decreased MAIT cells are associated with the occurrence of cGVHD and not a result of immunosuppressive treatments used to treat cGVHD.

Comparison of MR1 tetramer⁺ MAIT cells and CD161⁺TCR Vα7.2⁺MAIT cells in patients undergoing allogeneic HCT

The recent development of MR-1 tetramers permits the accurate identification of MAIT cells (6), although MR1 tetramers were not available when starting this study. Therefore, we evaluated associations between MR1: 5-OP-RU tetramer⁺ MAIT cells and MAIT cells defined by expression of CD161 and TCR Vα7.2 in CD3⁺TCR γδ⁺ cell populations in 65 patients undergoing allogeneic HCT (Fig. 3A). The proportion of MR1: 5-OP-RU tetramer⁺ MAIT cells among CD3⁺T cells was significantly associated with the proportion of MAIT cells defined by expression of CD161, TCR Vα7.2, and CD8⁺CD161⁺T cells but not CD4⁺CD161⁺T cells and MR1: 6-FP tetramer⁺ MAIT cells (Fig. 3B).

Next, we determined how many MR1: 5-OP-RU tetramer⁺ MAIT cells are in CD161⁺TCR Vα7.2⁺MAIT cells and its subpopulations based on CD4 and CD8 in 65 patients undergoing allogeneic HCT (Fig. 3C). MR1: 5-OP-RU tetramer⁺ MAIT cells

showed the mean of 35.6, 8.7, 50.1, 45.5, and 35.2% of CD161⁺TCR Vα7.2⁺, CD161⁺TCR Vα7.2⁺CD4⁺, CD161⁺TCR Vα7.2⁺CD8⁺, CD161⁺TCR Vα7.2⁺CD4⁺CD8⁺ (double positive), and CD161⁺TCR Vα7.2⁺CD4⁺CD8[−] (double negative) cells, respectively (Fig. 3D), indicating that CD161⁺TCR Vα7.2⁺CD4⁺ cells are not accurate in identifying MR1: 5-OP-RU tetramer⁺ MAIT cells.

Cytokine production and CD69 expression in CD161⁺T cells after HCT

To determine the functional capacity of each subset of circulating T cells after HCT, including CD8⁺CD161⁺T cells containing MAIT cells, we assessed alterations in production of granzyme B, IFN-γ, and IL-17 and expression levels of CD69 from each subset of T cells upon activation with PMA and IM in 38 patients after HCT (Fig. 4A, 4B). The proportions of IFN-γ⁺ of CD4⁺CD161⁺T cells and CD8⁺CD161⁺T cells were significantly lower in patients that received CBT compared with patients that received BMT in the unstimulated state (Fig. 4C). The proportions of CD69⁺ of CD4⁺CD161⁺T cells and CD8⁺CD161⁺T cells were significantly lower in patients that received CBT compared with patients that received BMT in the unstimulated state and under stimulation with PMA + IM (Fig. 4C). Levels of granzyme B in CD4⁺T cells and IL-17 in CD8⁺CD161⁺T cells were significantly lower in patients that received CBT compared with patients that received BMT (Fig. 4D). Expression of CD69 in CD4⁺T cells was also significantly lower in patients that received CBT compared with patients that received BMT (Fig. 4D). Contrastingly, there were no significant differences in proportions and expression levels in granzyme B, IFN-γ, and IL-17 between patients with or without cGVHD (Fig. 4E, 4F). Expression of CD69 in CD4⁺T cells was significantly higher in patients with cGVHD compared with patients without cGVHD (Fig. 4F).

We also evaluated alterations in the production of granzyme B, IFN-γ, and IL-17 and CD69 expression levels from each subset of T cells upon stimulation with IL-12, IL-15, and/or IL-18 in patients after CBT. Production of granzyme B, IFN-γ, and IL-17 and CD69 expression levels were elevated in CD8⁺CD161⁺T cells upon stimulation with IL-12, IL-15, and/or IL-18 (Supplemental Fig. 1B). Immunosuppressive treatment did not affect levels of granzyme B, IFN-γ, and IL-17 or CD69 expression from each subset of T cells upon activation with PMA and IM (Supplemental Fig. 1A). However, immunosuppressive treatment significantly affected granzyme B levels in CD4⁺CD161⁺T cells and CD8⁺CD161⁺T cells upon stimulation with IL-12, IL-15, and/or IL-18, but this effect was only observed in patients that received CBT (Supplemental Fig. 1C). Finally, SEA alone can also induce activation of MAIT cells in vitro (30). However, stimulation with SEA alone did not affect cytokine production or CD69 expression in each subset of T cells, regardless of cGVHD presence or immunosuppressive treatment administration (Supplemental Fig. 1D, 1E).

Alterations of transcription factor expression in CD161⁺T cells from patients with cGVHD after CBT

The expression profiles of transcription factors are associated with MAIT cell functions, including proapoptotic features (31), and are dependent on types of Ag presented by MR1 (32), origin (33), and distinct subset (34) of human MAIT cells. Therefore, we next investigated whether the transcription factor expression profile of

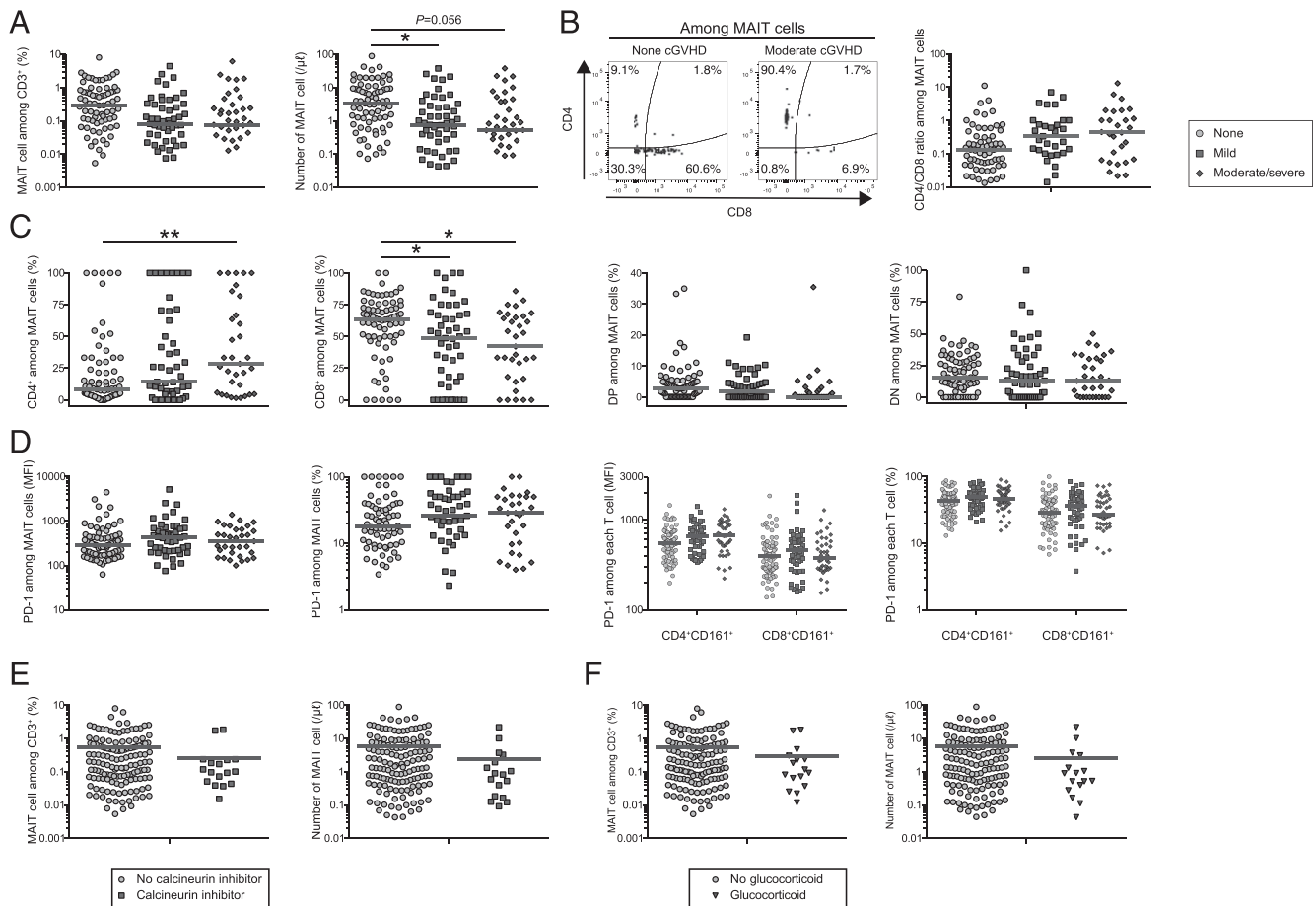


FIGURE 2. Circulating MAIT cells and their PD-1 expressions in long-term survivors after allogeneic HCT according to the NIH grade of cGVHD and administration of calcineurin inhibitors and glucocorticoids ($n = 173$). **(A)** Proportions and numbers of MAIT cells according to the NIH grade of cGVHD. **(B)** CD4/CD8 ratio among MAIT cells according to the NIH grade of cGVHD. **(C)** Proportions of CD4⁺, CD8⁺, CD4⁺CD8⁺ (double positive), and CD4[−]CD8[−] (double negative) among MAIT cells according to the NIH grade of cGVHD. **(D)** Expression levels and proportions of PD-1 among MAIT cells, CD4⁺CD161⁺T cells, and CD8⁺CD161⁺T cells according to the NIH grade of cGVHD. **(E)** Proportions and numbers of MAIT cells according to the administration of calcineurin inhibitors. **(F)** Proportions and numbers of MAIT cells according to the administration of glucocorticoids. The bar indicates mean values. Statistical significance was calculated by using unpaired Student *t* tests among two groups. Statistical significance was calculated by using one-way ANOVA followed by Bonferroni post hoc comparisons among three or more groups. * $p < 0.05$, ** $p < 0.01$.

a subset of T cells, including CD8⁺CD161⁺T cells, correlated with cGVHD after CBT. Under unstimulated conditions, there were no significant differences in expression of ROR γ t, PLZF, T-bet, and GATA3 in each subset of T cells between patients with and without cGVHD. Upon stimulation with SEA, the level of ROR γ t in CD4⁺T cells was significantly higher in patients with cGVHD compared with patients without cGVHD (Fig. 5A). Upon stimulation with IL-12, IL-15, and IL-18, the level of ROR γ t in CD8⁺CD161⁺T cells was significantly higher in patients with cGVHD compared with patients without cGVHD (Fig. 5A). However, we did not observe any association between expression levels of PLZF, T-bet, and GATA3 in each subset of T cells and cGVHD with different types of stimulation (Fig. 5B–D).

Association between gut microbiota and MAIT cell reconstitution after CBT

MAIT cells are not present in germ-free mice (7, 8), indicating that a commensal microbiota is necessary for normal development of MAIT cells. In addition, B cells are required for peripheral expansion of MAIT cells in mice and humans, as demonstrated in patients with mutated Bruton tyrosine kinase (3, 7). Therefore, we examined the short-term reconstitution of MAIT cells at 3, 6, and 12 mo in 27 patients undergoing CBT (Table II). Further, we

evaluated the effect of gut microbiota at 1 mo and circulating CD19⁺ B cells at 3 mo on MAIT cell reconstitution after CBT.

Reconstitution of MAIT cells occurred gradually after CBT. The number of MAIT cells at 12 mo was significantly higher than that at 3 mo after CBT (Fig. 6A). MAIT cell reconstitution was never observed in 18 (67%), 13 (48%), and 9 (33%) patients at 3, 6, and 12 mo after CBT, which was defined as $<0.01\%$. The proportions of PB CD19⁺B cells at 3 mo after CBT were significantly higher in patients with MAIT cell reconstitution at 6 and 12 mo after CBT compared with patients without MAIT cell reconstitution at each time point (Fig. 6B, 6C).

According to the presence and absence of MAIT cell reconstitution after CBT, the abundance and diversity of gut microbiota were evaluated at the family level (Fig. 6D). Principal coordinate analysis revealed a clear differentiation of gut microbiota between the presence and absence of MAIT cell reconstitution 12 mo after CBT (Fig. 6E). There were significant differences of unweighted UniFrac distances between the presence and absence of MAIT cell reconstitution at 12 mo after CBT (Fig. 6F). The α diversity of the microbiome, as measured by the Shannon index, was demonstrated to be significantly higher in patients with MAIT cell reconstitution 12 mo after CBT compared with patients without MAIT cell reconstitution (Fig. 6G). Consistently, the Chao1 index

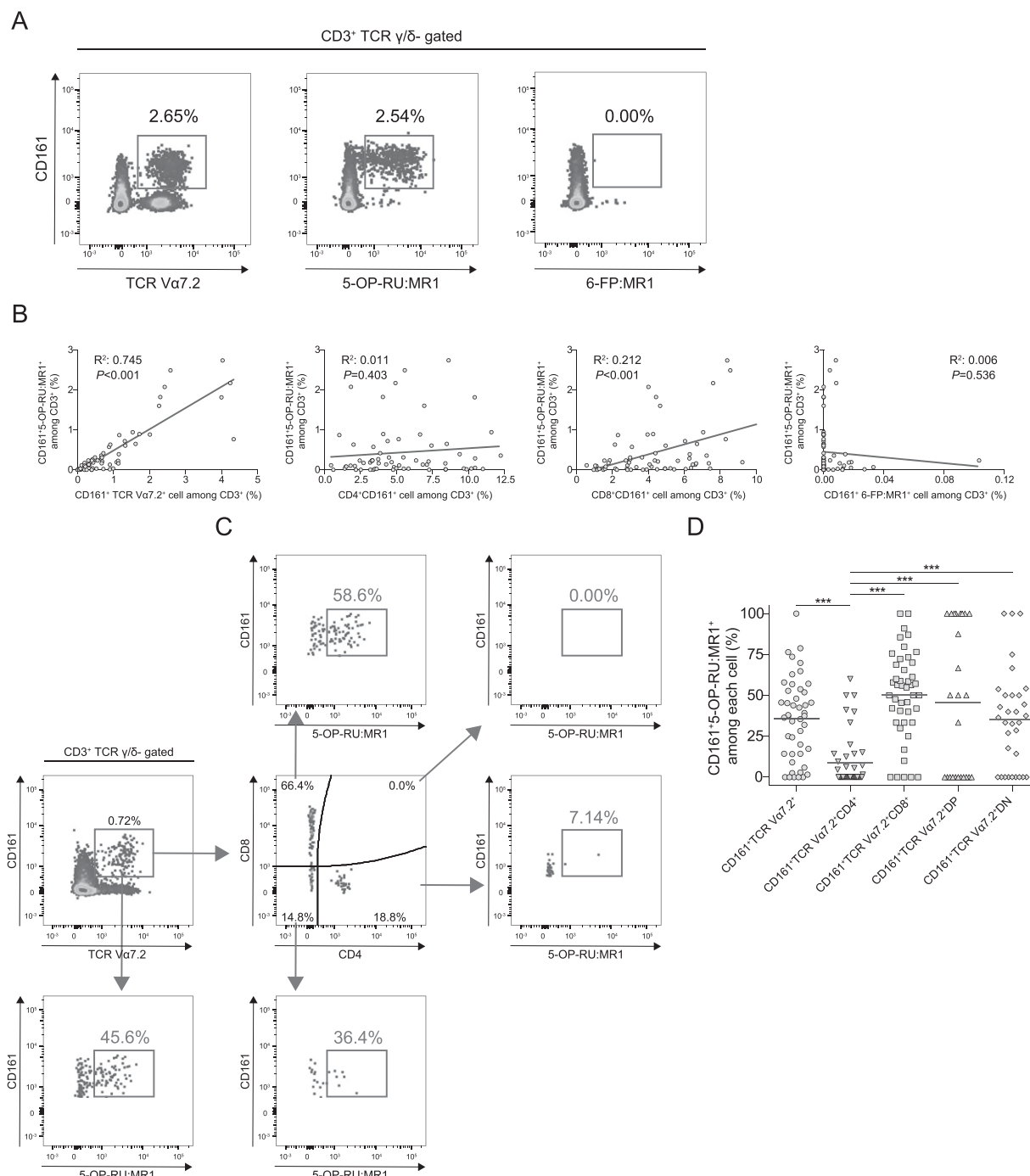


FIGURE 3. Comparison of MR1 tetramer⁺ MAIT cells and CD161⁺TCR Vα7.2⁺ MAIT cells in long-term survivors after allogeneic HCT. **(A)** Gating strategies for CD161⁺TCR Vα7.2⁺ MAIT cells and MR1: 5-OP-RU and 6-FP tetramer⁺ MAIT cells. **(B)** Correlation of proportion of MR1: 5-OP-RU tetramer⁺ MAIT cells, CD161⁺TCR Vα7.2⁺ MAIT cells, CD4⁺CD161⁺T cells, CD8⁺CD161⁺T cells, and MR1: 6-FP tetramer⁺ MAIT cells among CD3⁺ cells (n = 65). **(C)** Gating strategies for MR1: 5-OP-RU tetramer⁺ MAIT cells among CD161⁺TCR Vα7.2⁺, CD161⁺TCR Vα7.2⁺CD4⁺, CD161⁺TCR Vα7.2⁺CD8⁺, CD161⁺TCR Vα7.2⁺CD4⁺CD8⁺ (double positive), and CD161⁺TCR Vα7.2⁺CD4⁻CD8⁻ (double negative) cells. **(D)** Proportions of MR1: 5-OP-RU tetramer⁺ MAIT cells among each MAIT cell and its subpopulation based on CD4 and CD8 (n = 46). Correlations were examined using linear regression analysis. The bar indicates mean values. Statistical significance was calculated by using one-way ANOVA followed by Bonferroni post hoc comparisons among three or more groups. ***p < 0.001.

also had significantly higher diversity in microbiomes of patients with MAIT cell reconstitution 6 and 12 mo after CBT compared with patients without MAIT cell reconstitution (Fig. 6G). Although circulating MAIT cell reconstitution is reportedly correlated with increased gut abundance of *Blautia* species and *B. longum* in patients after allogeneic PBSCT (9), our data did not show any association between gut abundance of

these bacteria and MAIT cell reconstitution after CBT (data not shown).

Role of the gut microbiome putative riboflavin synthesis pathway in MAIT cell reconstitution after CBT

MAIT cell maturation requires commensal flora and products derived from riboflavin synthesis in these organisms. Therefore, we

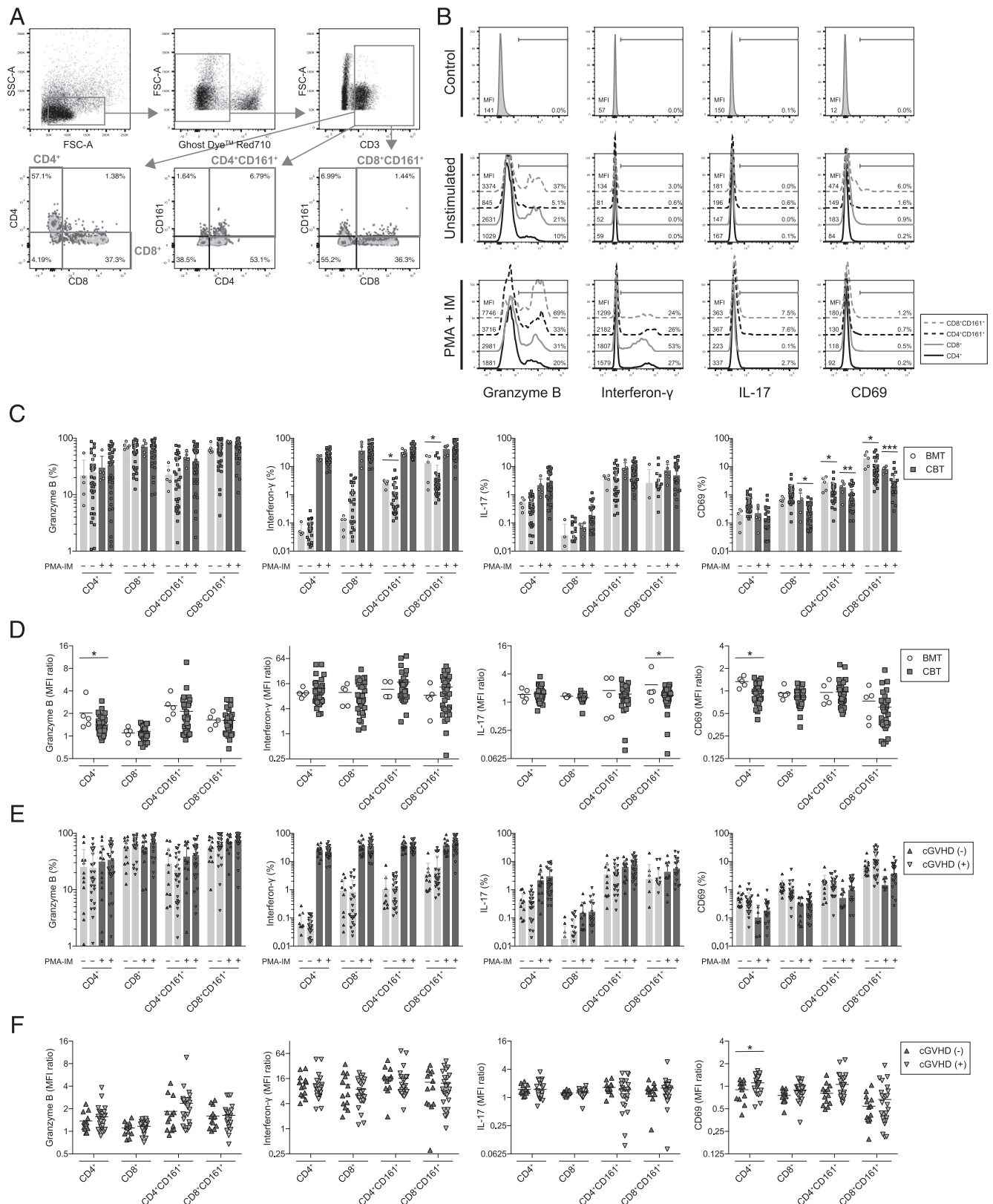


FIGURE 4. Cytokine production and CD69 expression in CD161⁺T cells after HCT under stimulation with PMA plus IM. **(A)** Gating strategies for CD161⁺T cells. **(B)** Representative histograms showing the expression levels of granzyme B, IFN- γ , IL-17, and CD69 in CD4⁺T cells, CD8⁺T cells, CD4⁺CD161⁺T cells, and CD8⁺CD161⁺T cells. **(C)** Proportions of granzyme B, IFN- γ , IL-17, and CD69 in each T cell subset according to donor source ($n = 38$). **(D)** Expression levels of granzyme B, IFN- γ , IL-17, and CD69 in each T cell subset according to donor source ($n = 38$). **(E)** Proportions of granzyme B, IFN- γ , IL-17, and CD69 in each T cell subset according to the NIH grade of cGVHD ($n = 38$). **(F)** Expression levels of granzyme B, IFN- γ , IL-17, and CD69 in each T cell subset according to the NIH grade of cGVHD ($n = 38$). Expression levels of granzyme B, IFN- γ , IL-17, and CD69 in each subset of T cells were evaluated by calculating the mean fluorescence intensity (MFI) ratio, which is defined as the MFI with stimulation relative to the MFI without stimulation. Data were expressed as the mean $\pm$ SD. The bar indicates mean values. Statistical significance was calculated by using unpaired Student t tests. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

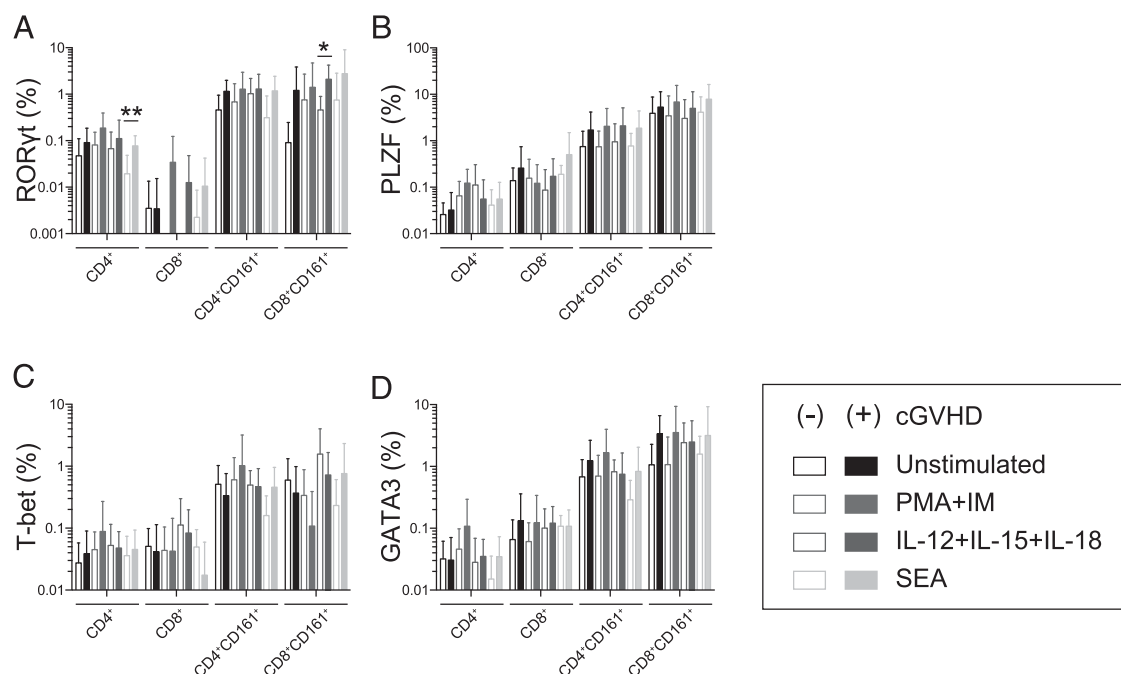


FIGURE 5. Expression of transcription factor in CD161⁺T cells after HCT under stimulation with PMA plus IM, IL-12, IL-15, and IL-18, and SEA. Proportions of (A) ROR γ t, (B) PLZF, (C) T-bet, and (D) GATA3 in each T cell subset according to the NIH grade of cGVHD in 24 patients with and without stimulation by PMA plus IM, IL-12, IL-15, and IL-18, and SEA. Data were expressed as the mean $\pm$ SD. Statistical significance was calculated by using unpaired Student *t* tests. **p* < 0.05, ***p* < 0.01.

used PICRUST to examine the associations between the putative riboflavin synthesis pathway in the gut microbiome at 1 mo and subsequent MAIT cell reconstitution after CBT. Among 10 riboflavin synthesis enzymes in the KEGG pathway, the amount of *ribB* (3,4-dihydroxy-2-butanone-4-phosphate synthase, RIB3) and *ribA* (GTP cyclohydrolase II, RIB1) genes was significantly higher in microbiomes of patients with MAIT cell reconstitution 6 and 12 mo after CBT compared with patients without MAIT cell reconstitution (Fig. 6H).

Discussion

Previous studies demonstrated that MAIT cell reconstitution was poor after allogeneic HCT (9, 15, 35). According to donor source, Bhattacharyya et al. (9) showed that MAIT cell reconstitution was poorer in adult patients who received double-unit CBT compared with patients who received PBSCT at all time points up to 1 y after HCT. However, the long-term reconstitution of MAIT cells between different donor sources after allogeneic HCT remained unclear. In our cross-sectional clinical study of long-term MAIT cell reconstitution, the number of MAIT cells correlated significantly with months after HCT in adult patients receiving CBT but not in patients receiving BMT/PBSCT. The cause of this difference remains unknown, but clonotype analysis revealed that MAIT cells present in HCT grafts might contribute to their reconstitution after proliferation is induced by TCR stimulation (9). Indeed, a previous study identified that cord blood grafts contained lower numbers of MAIT cells than those in adult donor sources, such as G-CSF-mobilized PB grafts for HCT (36). In contrast, our data also demonstrated that the number of MAIT cells increased to reach normal values observed in the age-matched healthy controls around 10 y after CBT, which is not consistent with a previous study by Ben Youssef et al. (35). They demonstrated that MAIT cells recovered toward normal values 5 y after CBT in children. Although the difference between these studies is unclear, this difference might be in part due to the fact that there was a marked delay in thymic recovery after CBT in adults

compared with those after CBT in children (37). Taken together, our findings suggest that reconstitution of MAIT cells after allogeneic HCT is dependent upon the type of donor source.

The ability of MAIT cells to recognize bacterial and fungal products presented by MR1 demonstrated a protective role in host immunity against infectious diseases (4, 11). However, MAIT cells are abundant in mucosal tissue and produce inflammatory cytokines following stimulation, so they therefore play a role in autoimmune disease. In clinical settings, a reduction or loss of circulating MAIT cells has been observed in patients with autoimmune diseases, such as multiple sclerosis (38, 39), rheumatoid arthritis (40), ankylosing spondylitis (41, 42), and type 1 diabetes

Table II. Characteristics of patients and CBTs

Characteristic	No. of Participants
Number of patients	27
Median age at CBT in y (range)	49 (17–67)
Sex, <i>n</i> (%)	
Male	13 (48.1)
Female	14 (51.9)
Disease type, <i>n</i> (%)	
AML	10 (37.0)
ALL	7 (25.9)
MDS	7 (25.9)
CML/MPN/CMML	3 (11.1)
Conditioning regimen, <i>n</i> (%)	
MAC	26 (96.3)
RIC	1 (3.7)
GVHD prophylaxis, <i>n</i> (%)	
CSP + MTX	17 (63.0)
CSP + MMF	6 (22.2)
CSP	4 (14.8)

ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CML, chronic myelogenous leukemia; CMML, chronic myelomonocytic leukemia; CSP, cyclosporine; MAC, myeloablative conditioning; MDS, myelodysplastic syndrome; MMF, mycophenolate mofetil; MPN, myeloproliferative neoplasm; MTX, methotrexate; RIC, reduced-intensity conditioning.

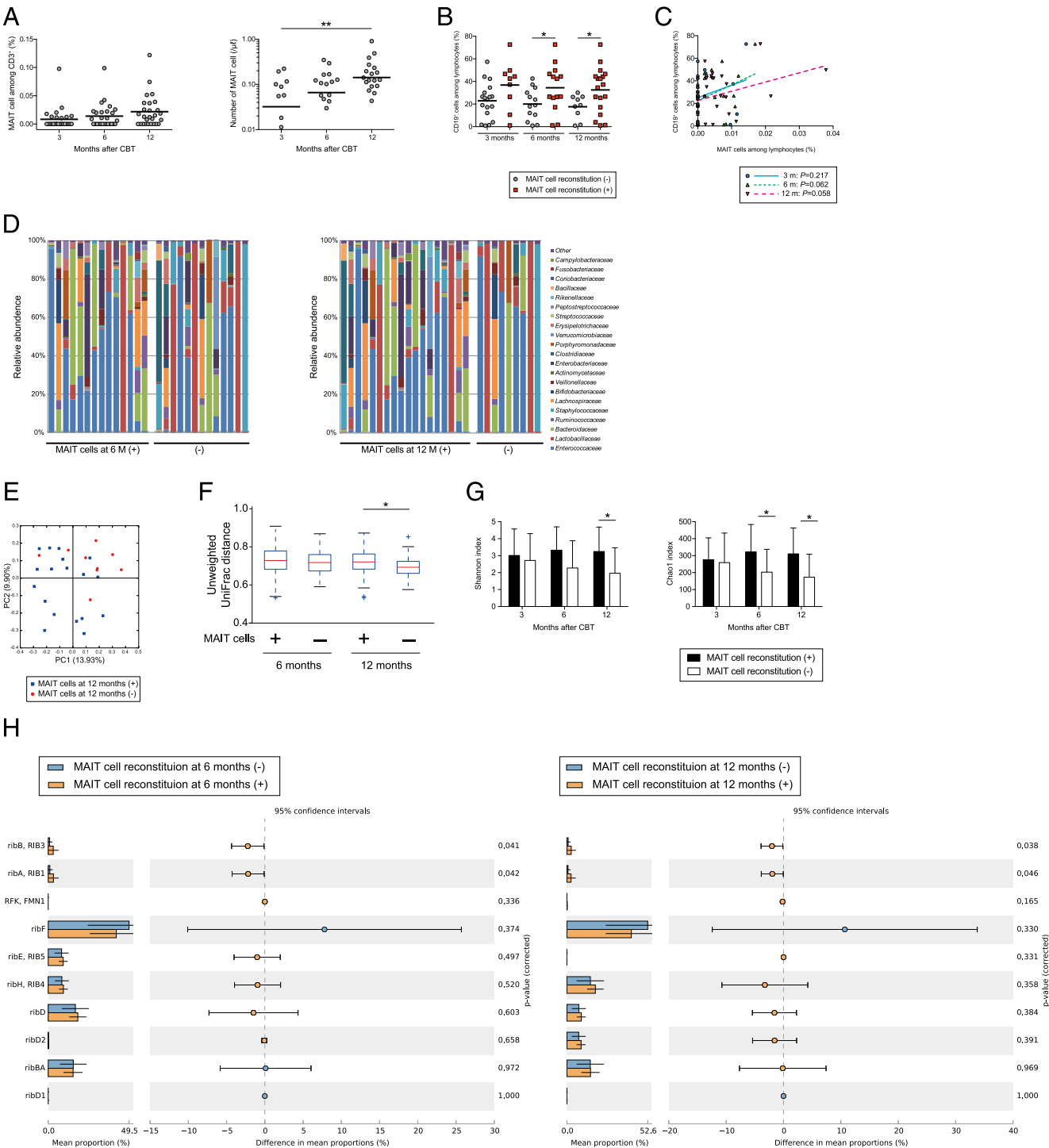


FIGURE 6. Reconstitution of MAIT cells, gut microbiota, and its putative riboflavin synthesis pathway in adult patients after CBT ($n = 27$). **(A)** Proportions and numbers of MAIT cells at 3, 6, and 12 mo after CBT. **(B)** Proportion of circulating CD19⁺B cells at 3 mo according to circulating MAIT cell reconstitution at 3, 6, and 12 mo after CBT. **(C)** Correlation between proportion of circulating CD19⁺B cells and MAIT cells at 3, 6, and 12 mo after CBT. **(D)** Family-level relative abundance of bacterial reads identified in individual stool samples according to circulating MAIT cell reconstitution at 6 and 12 mo after CBT. **(E)** Principal Coordinates Analysis of gut microbiome taxonomy according to the presence (blue) and absence (red) of circulating MAIT cell reconstitution at 12 mo after CBT. **(F)** Boxplots of unweighted UniFrac distances of gut bacterial operational taxonomic units according to circulating MAIT cell reconstitution at 6 and 12 mo after CBT. **(G)** The α diversity (Shannon index) and richness (Chao1 index) variation in gut microbiota according to circulating MAIT cell reconstitution at 3, 6, and 12 mo after CBT. **(H)** Mean proportion and difference in the mean proportion of predicted and significantly different inferences of riboflavin synthesis pathway inferences of gut microbiota in patients with (orange) or without (blue) subsequent reconstitution of circulating MAIT cells at 6 and 12 mo after CBT. The bar indicates mean values. Data were expressed as the mean $\pm$ SD. Statistical significance was calculated by using unpaired Student t tests among two groups. Statistical significance was calculated by using one-way ANOVA followed by Bonferroni post hoc comparisons among three or more groups. Correlations were examined using linear regression analysis. $*p < 0.05$, $**p < 0.01$.

(43). The reduction of circulating MAIT cells in autoimmune disease may be because these cells are recruited to lesions in the CNS in multiple sclerosis (39) or the synovial fluid in rheumatoid arthritis and ankylosing spondylitis (40–42). Cumulatively, these findings suggest that MAIT cells have the capacity to migrate and accumulate at inflamed sites in autoimmune disease.

Accordingly, previous studies demonstrated an association between decreased circulating MAIT cells and aGVHD in humans (9, 15, 16). Our data also demonstrate that decreased MAIT cells were significantly associated with development and severity of cGVHD, which is consistent with a previous report (17). Although Ben Youssef et al. (35) found that MAIT cells were not detectable in the intestines of pediatric CBT recipients that developed aGVHD, previous studies demonstrated an accumulation of CD161⁺T cells in the target organs of aGVHD and cGVHD in clinical biopsy samples (44, 45). However, we did not have biopsy samples available, so could not evaluate the correlation between infiltrated MAIT cells in target organs and decreased circulating MAIT cells in patients with cGVHD. Therefore, the dynamics of MAIT cell accumulation at the target sites of cGVHD could not be evaluated in the current study.

We also evaluated the function of circulating CD8⁺CD161⁺T cells in patients with cGVHD. Upon stimulation with IL-12, IL-15, and IL-18, the level of ROR γ t in circulating CD8⁺CD161⁺T cells was significantly higher in patients with cGVHD compared with patients without cGVHD. Activation of ROR γ t is essential for production of IL-17 in Th17 and Tc17 cells expressing CD161 (46), which could play a potential pathologic role in GVHD. However, Varelias et al. (14) reported that in mice, recipient-derived MAIT cells in the colon inhibit GVHD by attenuating the expansion of proinflammatory donor CD4⁺T cells. Bhattacharyya et al. (9) also demonstrated that human MAIT cells suppressed proliferation of CD4⁺T cells in vitro. These data supported the notion that MAIT cells could potentially inhibit GVHD. Therefore, the specific role of MAIT cells in the pathogenesis of GVHD will be elucidated in future investigations.

Our analysis of 27 patients within 1 y after single-unit CBT revealed that the number of MAIT cells gradually increased during this time period, but MAIT cell reconstitution was never observed up to 12 mo after CBT in 9 (33%) of 27 patients. MAIT cells undergo a three-stage intrathymic development pathway and continue to fully mature and expand in the periphery in humans and mice (8). However, the environmental factors important for circulating MAIT cell reconstitution after CBT remained unclear in clinical settings.

MAIT cell expansion requires normal microbiota, as MR-1–presented riboflavin synthesis products produced by the microbiota activate MAIT cells. The most potent MAIT cell agonists were unstable pyrimidine intermediates, including 5-OP-RU and 5-(2-oxoethylideneamino)-6-D-ribitylaminouracil (5-OE-RU), which are produced by nonenzymatic reactions between the riboflavin intermediate precursors 5-amino-6-D-ribitylaminouracil (5-A-RU) and methylglyoxal and glyoxal, respectively, and stabilized by binding to MR1 (5, 6). However, several studies showed that MAIT cells in human thymus, cord blood, and neonate PB do not bind the MR1: 5-OP-RU tetramer (8, 35), suggesting that gut microbiota is not required for early stages of MAIT cell development but is required for expansion in the periphery. Our data clearly demonstrated that the diversity of gut microbiota at 1 mo after CBT significantly affected subsequent MAIT cell reconstitution at 6 and 12 mo after CBT. Interestingly, PICRUSt analysis indicated that *ribB* and *ribA* expressed in gut microbiota were required for MAIT cell reconstitution after CBT. The riboflavin intermediate precursor 5-A-RU is converted from GTP by *ribA* and from ribulose-5-phosphate by *ribB* (47).

Our data also demonstrated that circulating B cells at 3 mo after CBT also significantly correlated with subsequent MAIT cell

reconstitution at 6 and 12 mo after CBT in a clinical setting, which is consistent with previous reports in mice and patients with mutated Bruton tyrosine kinase (3, 7). In addition, similar to previous reports that MAIT cells decrease with age (48, 49), recipient age also negatively correlated with MAIT cell numbers after allogeneic HCT in our study. Because older age negatively correlates with thymic output in patients after allogeneic HCT, thymic function may also affect MAIT cell reconstitution after allogeneic HCT in clinical settings.

Our study has several limitations that should be acknowledged. First, the main limitation of our study is a cross-sectional study. In fact, there were significant differences in the median time for sample collection between donor source, indicating that the difference in MAIT cell proportion and cytokine production between BMT and CBT might be in part due to the difference in the median time for sample collection between donor source. However, because disease recurrence and nonrelapse mortality are relatively common events after allogeneic HCT, it might be difficult to evaluate long-term MAIT cell reconstitution in a longitudinal study, and there were no longitudinal data for the evaluation of long-term MAIT cell reconstitution between donor source and its association with cGVHD. Therefore, further prospective longitudinal studies are needed to clarify associations between long-term MAIT cell reconstitution and its association with cGVHD. Second, MAIT cells have distinct cytokine secreting profiles and transcriptional profiles according to their peripheral locations, such as liver, gut, genital mucosa, lymphoid tissue, and PB (50–52). Therefore, the functional assay of circulating CD8⁺CD161⁺T cells in our study may not be fully indicative of immune function in the pathogenesis of cGVHD. Third, gut microbiota after allogeneic HCT can be dynamically changed by the conditioning regimen, administration of antibiotics, and altered mucosal integrity as well as age, sex, and race. The limitations of PICRUSt predictions must be considered in interpreting the data demonstrating an association between the riboflavin synthesis pathway in the gut microbiome and MAIT cell reconstitution after CBT. However, our results allowed us to assess gut microbiota regulation of circulating MAIT in a clinical setting.

In conclusion, long-term MAIT cell reconstitution was poorer in patients who received CBT compared with patients who received BMT/PBSCT, suggesting that long-term MAIT cell reconstitution is dependent on the type of donor source. Our findings evaluated MAIT cell reconstitution and a potential correlation with the riboflavin synthesis pathway of gut microbiota after CBT. This finding highlights an important role for interactions between circulating MAIT cells and gut microbiota in humans. Together with accumulating evidence for the role of circulating MAIT cells and CD8⁺CD161⁺T cells in cGVHD, our findings suggest that a decrease in MAIT cells may be a biomarker and potential therapeutic target for cGVHD.

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Disclosures

The authors have no financial conflicts of interest.

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